

4) Diferencia de un polinomio de Bernoulli.

$$\Delta \hat{B}_v^{\wedge-1}(x) = v \hat{B}_{v-1}^{\wedge-1}(x)$$

$$\Delta \left[\frac{\tau^{\wedge} e^{\tau x}}{(\tau^{\wedge}-1)^{\wedge}} \right] = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} \Delta \left(\hat{B}_v^{\wedge}(x) \right)$$

$$\frac{\tau^{\wedge}}{(\tau^{\wedge}-1)^{\wedge}} (e^{\tau x+1} - e^{\tau x}) = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} \Delta \left(\hat{B}_v^{\wedge}(x) \right)$$

$$\frac{\tau^{\wedge}}{(\tau^{\wedge}-1)^{\wedge}} e^{\tau x} (\tau^{\wedge}-1) = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} \Delta \left(\hat{B}_v^{\wedge}(x) \right)$$

$$\tau \cdot \frac{\tau^{\wedge-1}}{(\tau^{\wedge}-1)^{\wedge-1}} e^{\tau x} = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} \Delta \left(\hat{B}_v^{\wedge}(x) \right)$$

$$\tau \cdot \sum_{v=0}^{\infty} \frac{\tau^v}{v!} \hat{B}_v^{\wedge-1}(x) = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} \Delta \left(\hat{B}_v^{\wedge}(x) \right)$$

$$\sum_{v=0}^{\infty} \frac{\tau^{v+1}}{(v+1)!} (v+1) \hat{B}_v^{\wedge-1}(x) = \frac{\tau^0}{v_0} \underbrace{\Delta \left(\hat{B}_0^{\wedge}(x) \right)}_{=0} + \sum_{v=1}^{\infty} \frac{\tau^v}{v!} \Delta \left(\hat{B}_v^{\wedge}(x) \right)$$

$$\sum_{v=1}^{\infty} \frac{\tau^v}{v!} \cdot v \hat{B}_{v-1}^{\wedge-1}(x) = 0 + \sum_{v=1}^{\infty} \frac{\tau^v}{v!} \Delta \left(\hat{B}_v^{\wedge}(x) \right)$$

Luego

$$v \hat{B}_{v-1}^{\wedge-1}(x) = \Delta \hat{B}_v^{\wedge}(x)$$